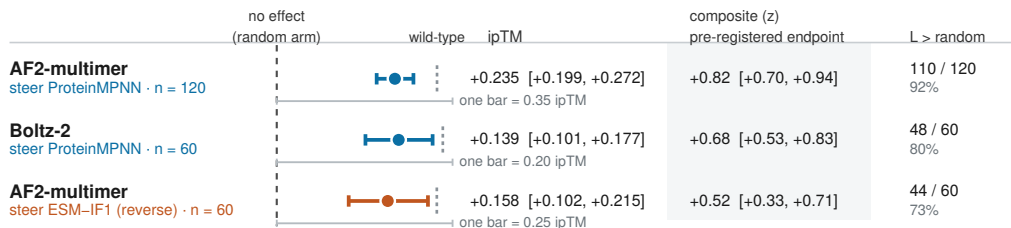


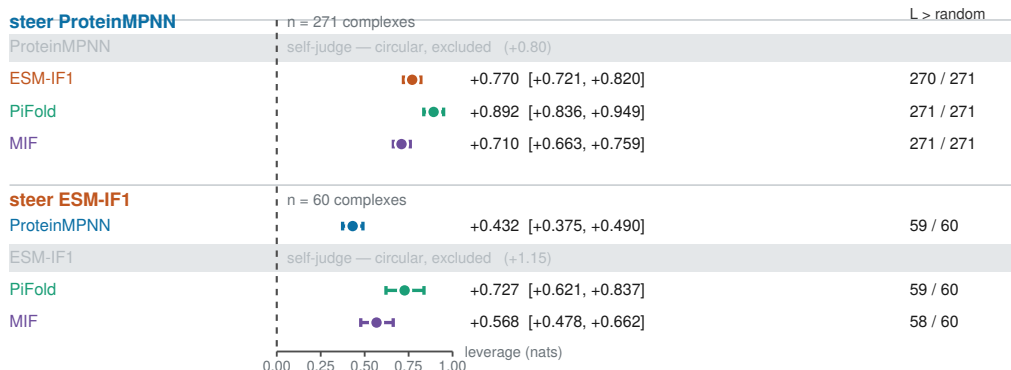
a a second folder, a second steered model, the same direction
paired Δ (L – random) per complex, mean over $k = 3$ · complex-clustered 95% CI · $P(>0) = 1.00$, every row



each row is read in its OWN calibration — every interval clears zero, but the three magnitudes are NOT comparable
on each row the tilt covers 74%, 73%, 69% of the random → wild-type distance — the comparison the raw Δ cannot make

b and under four judge architectures — never the one that was steered

paired Δ judge-leverage (L – random), $\alpha = 2$ · self-judged diagonal excluded · $P(>0) = 1.00$ on all six



every non-self cell: L > random in $\geq 97\%$ of complexes — four architectures, both directions, never the steered model
all six share ONE leverage scale — unlike a, these cells are one instrument family, so lengths do compare

a, paired Δ per complex, mean over $k = 3$ folds; complex-clustered bootstrap, 5,000 replicates, seed 20260803. Composite = z-mean of (ipTM, -interface pAE, interface pLDDT), z-scored WITHIN each run ($n = 117 / 58 / 58$). The 60 Boltz-2 and the 60 reverse-direction complexes are the SAME set, nested in the 120; on those 60, AF2 / steer-ProteinMPNN gives +0.226 (70% of its span), so the AF2 / Boltz-2 difference is not a complex-set effect. b, judge-leverage = the per-complex mean L over the sampled interface residues; method dumped3, same bootstrap.